



Question Set

Question 1

Consider a European equity option under geometric Brownian motion. Define the state variables (spot, strike, rate, dividend yield, volatility, maturity, and Greeks) and derive the Black-Scholes PDE and closed-form call value. State the measure, filtration, and tradability assumptions.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

Question 2

For a European equity option under geometric Brownian motion, explain which probability measure is natural for pricing. Write the density-process logic needed if the starting model is under the physical measure.

Question 3

Identify the discounted tradable or density-adjusted process in a European equity option under geometric Brownian motion. Prove the martingale condition and name one modelling choice that would break it.

Question 4

Using spot 100, strike 100, rate 4 percent, dividend 1 percent, volatility 22 percent, maturity one year, construct a two-step numerical calculation. Show intermediate state values before discounting or aggregating.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

Question 5

For a European equity option under geometric Brownian motion, compute and interpret delta, gamma, theta, vega, and PDE residual. Explain the hedge sign and the unit of each sensitivity.

Question 6

Suppose the desk observes implied volatility from a quoted option premium. Write the calibration objective, the weights you would use, and the parameter-identifiability checks.

Question 7

State the terminal and boundary conditions for a European equity option under geometric Brownian motion. Explain how terminal payoff and low/high spot asymptotics prevents economically impossible prices.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

Question 8

Design a Monte Carlo or lattice estimator for a European equity option under geometric Brownian motion. Specify risk-neutral GBM paths benchmarked to the closed form, the estimator, the discounting rule, and the standard error.

Question 9

A junior quant implements a European equity option under geometric Brownian motion and gets a stable but wrong answer. Explain how using physical drift in an arbitrage-free price creates the error and design a diagnostic test.

Question 10

Write an NumPy algorithm for a European equity option under geometric Brownian motion. Include inputs, outputs, diagnostics, and one benchmark case.

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Question 11

Starting from the payoff definition of a European equity option under geometric Brownian motion, derive a replication, dynamic-programming, or conditional-expectation representation without skipping the economic argument.

Question 12

Create a stressed-market version of spot 100, strike 100, rate 4 percent, dividend 1 percent, volatility 22 percent, maturity one year. Recompute the value or risk metric and explain which term dominates the change.

Question 13

Use a second-order Taylor approximation for a European equity option under geometric Brownian motion. Separate linear exposure, convexity exposure, cross terms if relevant, and residual model risk.

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Question 14

Show how a change of numeraire or change of measure would simplify one expectation in a European equity option under geometric Brownian motion. Identify what becomes a martingale after the change.

Question 15

Give a proof-style answer showing when the wealth process for a European equity option under geometric Brownian motion is self-financing. Explain why predictable trading strategies matter.

Question 16

Market quotes for implied volatility from a quoted option premium are noisy. Propose a robust calibration workflow with initial guesses, constraints, residual plots, and out-of-sample checks.

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Question 17

For a European equity option under geometric Brownian motion, compare the mathematical boundary condition with the trader's intuition. Give one limiting case where the answer should be obvious.

Question 18

Design a convergence study for risk-neutral GBM paths benchmarked to the closed form. Specify path counts, time-step refinement, confidence intervals, and the exact quantity plotted.

Question 19

Find one way that using physical drift in an arbitrage-free price could pass a superficial unit test. Write a stronger test that would fail for the biased implementation.

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Question 20

Explain a European equity option under geometric Brownian motion and code the core routine. Give the derivation outline, pseudocode, and three numerical sanity checks.